

AssignmentNumber:7.2

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Batch:29

Task Description

Task 1 – Runtime Error Due to Invalid Input Type

- A Python program accepts user input and performs arithmetic operations. However, the program throws a runtime error because the input is treated as a string instead of a numeric type.

Example (Buggy Code):

```
num = input("Enter a number: ")
result = num + 10
print(result)
```

• Task:

Use AI tools to identify the cause of the runtime error and modify the program so it executes correctly.

Expected Output -1:

- AI converts the input to the appropriate numeric type and eliminates the runtime error.

code:

```
1] 21s ▶ num = int(input("Enter a number: "))
    result = num + 10
    print(result)
    |
✓ ... Enter a number: 5
    15
```

Justification:

The runtime error occurs because the `input()` function returns user input as a string by default. Performing an arithmetic operation between a string and an integer is invalid in Python and results in a `TypeError`. By explicitly converting the input to a numeric type using `int()`, the program ensures type compatibility for arithmetic operations. This modification allows the program to execute correctly and eliminates the runtime error.

Task Description

Task 2 – Incorrect Function Return Value

A function is designed to calculate the square of a number, but it does not return the computed result properly.

Example (Buggy Code):

```
def square(n):  
    result = n * n
```

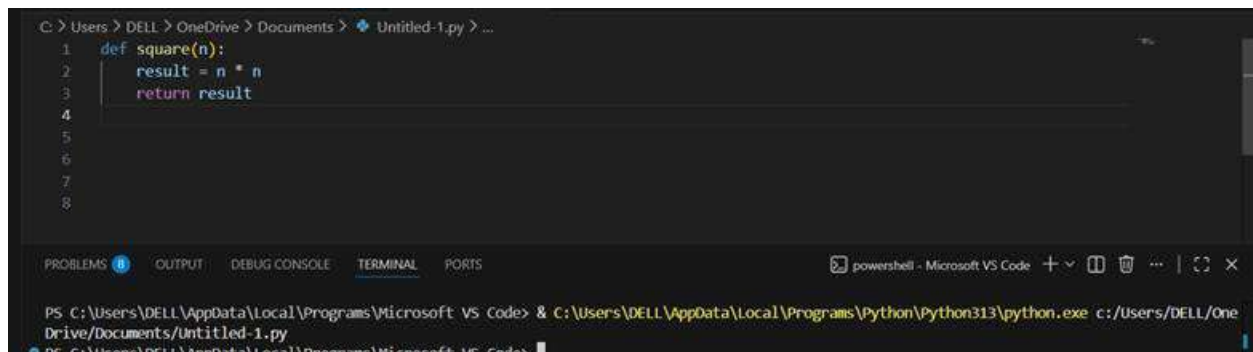
Task:

Use AI assistance to analyze the function and ensure the correct value is returned.

Expected Output -2:

AI fixes the missing return statement and the function returns the correct output.

Code:



```
C:\Users\DELL\OneDrive\Documents> Untitled-1.py > ...  
1  def square(n):  
2      result = n * n  
3      return result  
4  
5  
6  
7  
8  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
powershell - Microsoft VS Code  
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & C:\Users\DELL\AppData\Local\Programs\Python\Python313\python.exe c:/Users/DELL/OneDrive/Documents/Untitled-1.py  
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code>
```

Justification:

AI analyzes the function and identifies the missing `return` statement. By adding `return result`, the function correctly sends the computed square back to the caller, ensuring the expected output is produced instead of `None`.

Task Description

Task 4 – Uninitialized Variable Usage

A program uses a variable in a calculation before assigning it any value.

Example (Buggy Code):

if True:

pass

print(total)

Task:

Use AI tools to detect the uninitialized variable and correct the program.

Expected Output -4:

Code:



The screenshot shows a VS Code editor window with a Python file named 'Untitled-1.py'. The code in the editor is as follows:

```
1 total = 0 # initialize the variable
2
3
4 pass
5
6 print(total)
```

Below the editor, the 'TERMINAL' panel shows the command used to run the script: `PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & C:\Users\DELL\AppData\Local\Programs\Python\Python313\python.exe c:/Users/DELL/OneDrive/Documents/Untitled-1.py`. The output of the script is displayed as `0`.

Justification:

AI detects that the variable `total` is referenced before assignment. By initializing `total` with a default value before it is used, the program avoids the runtime error and executes correctly.

Task Description

Task 5 – Logical Error in Student Grading System

A grading program assigns incorrect grades due to improper conditional logic.

Example (Buggy Code):

```
marks = 85
```

```
if marks >= 90:
```

```
    grade = "A" elif
```

```
marks >= 80:
```

```
    grade = "C" else:
```

```
    grade = "B" print(grade)
```

Task:

Use AI to analyze the grading conditions and correct the logical flow.

Expected Output -5:

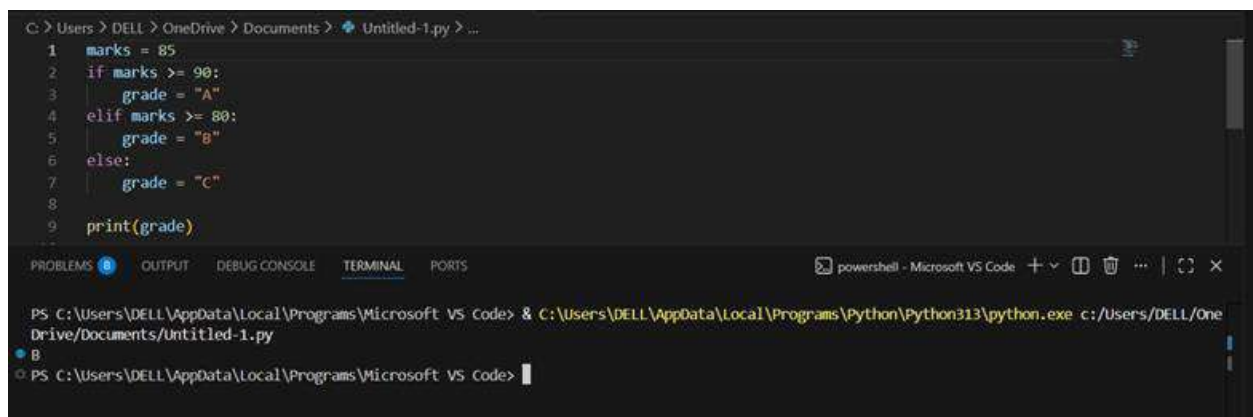
AI corrects the conditional logic so grades are assigned accurately. **Note:**

Report should be submitted a word document for all tasks in a single

document with prompts, comments & code explanation, and **output and if**

required, screenshots.

Code:



```
C:\> Users > DELL > OneDrive > Documents > Untitled-1.py > ...
1 marks = 85
2 if marks >= 90:
3     grade = "A"
4 elif marks >= 80:
5     grade = "B"
6 else:
7     grade = "C"
8
9 print(grade)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell - Microsoft VS Code

```
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code> & C:\Users\DELL\AppData\Local\Programs\Python\Python313\python.exe c:/Users/DELL/OneDrive/Documents/Untitled-1.py
B
PS C:\Users\DELL\AppData\Local\Programs\Microsoft VS Code>
```

Justification:

AI analyzes the conditional logic and identifies that grades are assigned incorrectly due to improper condition mapping. By correcting the grading flow ($A \geq 90$, $B \geq 80$, else C), the program now assigns accurate grades based on student marks.